Course Title & Course Code: Discrete Structure & Graph Theory (3IT02)

Session: 2024-2025 (Autumn)

Class: Second Year / **Semester:** III

Name of the Course Teacher: Dr. S. D. Padiya

Title of the Innovative Practice: Flipped Classroom

Objectives/ Goals of the Practice:

- To provide greater access to quality educational resources.
- To implement flipped classroom and e-learning methodologies.
- To support self-paced and self-directed learning.
- To motivate students through multimedia engagement.
- To enhance understanding and retention of complex concepts in DS&GT.

YouTube Channel Overview:

- Channel Name: Sagar Padiya
- Number of Videos Uploaded: 27
- Content Coverage: Complete syllabus of Discrete Structure & Graph Theory (3IT02)
- Video Format: Lecture-style explanations, problem-solving demonstrations, concept visualizations, and real-life applications.
- Channel Link: https://www.youtube.com/@SagarPadiya

Methodology

1. ***Flipped Classroom Approach:***

   Students are assigned video lectures as preparatory material before attending class. In-class time is then dedicated to active learning through discussions, problem-solving, quizzes, and collaborative activities. This model helps:

   - Maximize classroom engagement.
   - Encourage deeper conceptual understanding.
   - Foster peer-to-peer learning.

2. ***E-Based Learning:***

   The YouTube videos serve as a reliable, on-demand educational resource. Students can revisit the content anytime to revise or clarify doubts. This digital repository supports:

   - Flexible and personalized learning.

- Access beyond geographical and time constraints.
- Continuous learning support for slow and fast learners alike.

3. ***Educational Video Integration:***

   Each video is designed to:

   - Simplify complex topics using animations and examples.
   - Facilitate motivation through interactive content.
   - Encourage inquiry-based learning with thought-provoking questions and challenges.

Impact and Outcomes:

1. Student Feedback: Initial feedback indicates increased comprehension and satisfaction. Students appreciate the ability to learn at their own pace.
2. Academic Performance: Preliminary results show improved performance in assessments, especially in topics reinforced by video content.
3. Classroom Engagement: Enhanced student participation in classroom activities due to prior exposure to video content.

Advantages of the Initiative:

1. Cost-effective and scalable educational delivery.
2. Reusability of content across academic years and batches.
3. Encouragement for blended learning culture in the institution.
4. Support for inclusive education, catering to diverse learning needs.

POs Mapped: PO1, PO2, PO3, PO4, PO12

Reflective Critique:

- The videos offer students flexible learning opportunities, allowing them to learn at their own pace and revisit complex topics as needed. This is particularly beneficial for students who may struggle with in-class learning due to varied learning speeds or external constraints.

  Reflection: Though flexibility was increased, not all students may possess the intrinsic motivation required for self-paced learning. Strategies to ensure engagement, such as integrated quizzes or learning analytics, could be explored.

- The video content is comprehensive and well-aligned with the syllabus. However, the pedagogical design can always be enhanced by incorporating interactive elements, real-world applications, or brief assessment checkpoints within or after each video.
  Reflection: More attention could be given to designing the videos based on cognitive load theory and multimedia learning principles to maximize understanding and retention.

- Flipping the classroom has increased student participation during physical classes. However, tracking whether students are genuinely watching and understanding the videos remains a challenge.
  Reflection: Embedding simple formative assessments or using platforms like EdPuzzle or Google Forms linked with the videos could provide useful feedback on student engagement and comprehension.

- The multimedia approach is motivating for many students. Nevertheless, some students may lack the self-discipline or time-management skills required for independent study.
  Reflection: This highlights the need to provide students with guidance on how to effectively use video content and manage their learning schedules.

Photos of the Activity

Challenges and Recommendations

1. ***Challenges:***
   - Ensuring consistent student engagement with the videos.
   - Need for regular content updates and quality enhancement.
   - Limited analytics on viewer behaviour and effectiveness.
2. ***Recommendations:***
   - Integrate video analytics tools to monitor engagement and learning patterns.
   - Encourage students to contribute by submitting questions or feedback on video topics.
   - Organize periodic live Q&A sessions or discussions based on video content.